

Meta-interactivity and Playful Approaches for Musical Composition

Mário Escarce Junior

m.escarce@lancaster.ac.uk

Lancaster University

Lancaster, Lancashire, UK

ABSTRACT

The concept of gamified interactive models and its novel extensions, such as playification, has been widely approached in order to engage users in many fields. In fields such as HCI and AI, however, these approaches were not yet employed for supporting users to create different forms of artworks, like a musical corpus. While allowing novel forms of interactivity with partially-autonomous systems, these techniques could also foster the emergence of artworks not limited to experts. Hence, in this research we introduce the concept of meta-interactivity for compositional interfaces, which extends an individual's capabilities by the translation of an effort into a proficiency. We study how this approach can be effective through the development of novel systems that enables novices to compose coherent musical pieces through the use of imagetic elements in a virtual environment.

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1 CURRENT STATUS IN THE PHD PROGRAM

Currently I am a 2nd year Computer Science PhD student at Lancaster University. I started in the program in April 2020, and my target end date is March of 2023. Lancaster University's Computer Science PhD Program lengths 3 years, with the possibility of an extension for one extra year. During this time, PhD candidates must go through several appraisals until the viva presentation. More specifically, there are 6 appraisals in total. Of these, the 10-month appraisal is the confirmation panel that defines whether or not the student can proceed in the program. The appraisals have a fixed panel of subject specialists and a member from another department, and this format follows until the viva presentation. I have currently just passed the 16-month appraisal.

2 RESEARCH OUTLINE

We are currently focused on the procedural generation of music. Emergent music is a widely discussed subject and a very promising field, with many conferences increasingly approaching the theme.

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Allowing users with different backgrounds and expertise levels to compose a coherent musical corpus establishes many challenges, and our commitment to overcoming them through the conception of novel forms of human-system collaboration is inspiring us in the development of the systems such as Bubble Sounds, which we will briefly present further on.

In short, our focus is on designing partially autonomous models capable to allow users to achieve complex tasks, such as music composition and performance, in a more ludic and engaging way. We call it "partially" autonomous because many works related to procedural content generation are actually self-sufficient on the way they process information and provide its outcome. For instance, Machine Learning models can successfully achieve the generation of artworks, but usually through the following linear workflow: (i) They demand training under a specific guideline in order for the system to learn a determined task (e.g. drawing on a canvas, composing a melodic line, etc); (ii) process/render the information in a self-sufficient fashion; (iii) provides a "ready-made" outcome.

On the other hand, as previously mentioned, the approaches we intend on proposing are heavily based on the collaboration between a user and a system. We assume that it is very difficult to neglect the human factor in this kind of human-system relation, as any form of traditional artistic expression. With this in mind, our focus will be in transforming, on a collaborative-basis, an expressive motivation of an individual into a coherent outcome that might be acknowledged by the general public as a human-made piece. In this way, our work focuses not only on providing new forms of expression through interactive systems, but also in maximizing the user's performance on a giving task in a virtual world without excessively limiting him/her on their creative process.

Thus, in this research we intend to introduce the concept of meta-interactivity, and how it can establish a relation with different expressive endeavors in order to homogeneously produce a desired outcome. This approach can be seen as a powerful gamified tool that can be used to explore a user's creative proficiencies, such as drawing, and extend its potential to other expressive fields, such as music production. To explore this concept, we will develop and present novel interactive systems that enables non-experts in musical theory and performance to produce an original and pleasant ambient music piece through a friendly interface that allows him/her to use visual elements, such as colors, as tools to sculpt a coherent musical corpus.

3 RELATED WORK

This is an interdisciplinary work, that benefits from, collaborates and dialogue with different fields. For instance, HCI, AI, game studies, procedural content generation and fine arts are some that

provides us fertile soil to discuss the many elements that orbits the meta-interactive approach. However, we highlight that the main contribution of this work, especially regarding the system we will present, resides in the following fields: (i) Virtual Reality Musical Instruments (VRMIs) and Immersive Virtual Musical Instruments (IVMIs), as discussed by Serafin et al. and Berthaut et al. [2, 24]; (ii) studies in interfaces capable to foster efficient devices for the collaboration between human and synthetic agents.

Video game industry. More specifically in the music video games genre, we can find works that explore different forms of interactivity through the collaboration between musical systems and a human agent. E.g., *Vib Ribbon* [17] offers a gameplay mechanic where the player interacts with rhythmic obstacles generated by the music that is being played. Similarly, *Patapon* [15] proposes the user to execute rhythmic sequences on the joystick in order for the characters to proceed across levels. In these examples, however, the space for the user expressiveness is very restricted, since, in the first, the player does not control the music that is being generated. In the other two examples the only action allowed is usually to press determined buttons at predetermined moments.

HCI approaches. Interactive design approaches aims to bring daily life experiences into the virtual environment. For instance, the *Drift Table* [10] is an electronic coffee table that displays slowly moving aerial photography controlled by the distribution of weight on its surface. It was designed to instigate reflection on the ways technology can support ludic activities, as well as improve our general understanding of ludic design. *SoundSelf: A Technodelic* [9] provides the user a guided meditation experience. Through the usage of voice input, it is possible to explore a sound-based virtual world where modulations of the user's voice creates the feeling of presence in an abstract environment. Previous works also explore possibilities for new musical practices to emerge through novel mechanic design. For instance, Koray et al. [26] use the notion of "cultural probe" – a technique used to gather inspirational ideas from individuals' lives – as input to digital musical instruments. Regarding the production of interactive interfaces for the emergence of music, *Electronauts* [25] gives the users the opportunity to become a DJ in a VR experience, where it is possible to jam with other users and even professional musicians through the remixing of pre-existing tracks. In music and interactivity, Holland et al. [11] introduce recent research in Music and Human Computer Interaction, also known as Music Interaction, and discusses its role in HCI. According to the authors, current musical activities include a wide range of expressive forms, besides performance and composition; it also includes collaborative music making and human and machine improvisation. From this perspective, specific issues may arise, including: (i) whether Music Interaction should be easy for newcomers; (ii) what can be learned from the experience from those "in the groove", that is, users that already find themselves in a flow of the experience, according to Mihaly Csikszentmihalyi's definition of immersion [5]; and (iii) what can be learned from the commitment of non-experts in musical theory and practice with the system mechanics and interface. In performance, *Astaire* [28] proposes a collaborative hybrid VR dance game for two players sharing the same environment.

Many HCI works also seek a better understanding of how music impacts players behavior in their experience while playing games.

For instance, Rogers et al. [22] investigate whether it is possible to influence video game players on risk-taking behaviour through the presence of elements in the background music. The authors concluded that music plays a major role in the immersion of players, as well as in their learning process of the gameplay mechanics. Other works also assess how much music in games contributes to the player experience, more specifically considering VR approaches [23].

Virtual Reality Musical Instruments (VRMIs) and Immersive Virtual Musical Instruments (IVMIs). Works have been exploring different forms in which users interact with virtual musical instruments. For instance, Mäki-Patola et al. [16] introduce and analyze four gesture-controlled musical instruments designed to allow for rapid experimentation of new interfaces and control mappings. Cabral et al. [3] present a novel 3D virtual instrument with a particular mapping of touchable virtual spheres to notes and chords of a given musical scale. The objects are metaphors for note keys organized in multiple lines, forming a playable spatial instrument where the player can perform certain sequences of notes and chords across the scale employing specific gestures. Mitchusson [18] uses the audio module and the Virtual Reality capabilities from the Unity engine to generate, through a dice-rolling mechanic, sample effects and audio mixing to generate experimental music outcomes.

AI approaches. A common approach is to use Deep Learning models to support the generation of music and new composition methods. E.g., Roberts et al. [21] enable the integration of generative models through the use of the Google Magenta Interface to arouse musicians creativity. This system was experimented on a live jazz performance with a piano and drums duet. Also, Huang and Raymond [12] create coherent music with melodic and harmonic structures that can be acknowledged as pieces composed by a human agent. Castro [4] trained Deep Learning models to perform coherent improvisations. Agarwala et al. [1] use generative Recurrent Neural Networks to create musical sheets without predefining compositional rules to the models. Rocksmith [27] presents a mode where it is possible to explore the cooperation between the user and an AI system through a conventional musical instrument, such as an electric guitar. The user can jam with the system, which provides a non-adaptive background track (i.e., it is the user who must adapt to the tempo of the song).

Still AI related, but more specifically in the Generative Art field, many researchers develop AI systems where humans and algorithms cooperate in art emergence. We can observe very inspiring works that foster different forms of interaction such as performance and improvisation through the interaction with partially autonomous systems [8, 19, 20]. For instance, Davis et al. [6] propose an experience where a user takes turn with a computer AI system when drawing in the same canvas. In Jacob and Magerko [13], a human and an agent collaborate to produce movement-based performance pieces using a co-creative agent. These works, however, happen apart from the electronic games development efforts, thus are not focused on providing friendly imagetic interfaces that guides non-experts in any kind of content generation process, demanding them to be properly trained in order for the desired outcome to emerge.

In general, many works discussed in this section explore a wide variety of approaches from HCI to collaboration between AI systems and human agents. They do, however, demand the user proper training in order for the generative outcome to be “coherent”. Also, some examples only allow users to edit pre-existing tracks, not actually allowing users to compose music from scratch. In our work, however, we propose an approach where even non-expert users can compose a music piece that can be acknowledged by the general public as something created by a trained musician.

4 META-INTERACTIVITY

In this research we introduce the concept of meta-interactivity as a novel approach to be explored in gamified experiences, HCI and AI systems. With this approach we aim to engage users to accomplish complex tasks, such as music composition, through a ludic effort, that not only makes the tasks easier to be accomplished but also more engaging and fun. The way we propose such effect is rather different from other approaches, such as gamification [7], for example. Our effort is centered on algorithms that establish bridges capable to translate different artistic endeavors, that are mainly based on expressive efforts, into new artistic instances. From this novel form of user-system relation, we expect coherent forms of artworks emerging from interfaces designed with ludic elements, that evokes an aptitude of the user and converts it into a new instance. It can be comprehended as a way to translate different creative motivations, like “drawing” to “musical composition”, in order to propose different relations between the system and the users, such as “painting a music”, “playing” (in the musical sense) on a canvas. Ultimately, it consists of designing minimalist, very intuitive, and easy-to-use interactive models that generate complex outcomes.

4.1 Bubble Sounds

Bubble Sounds is an interactive musical system we developed as an artistic installation, and that was also used as a proof of concept to verify the efficiency of the meta-interactive approach. The system offers a game-like experience, but its proposal is rather different from the common entertainment-based approach focused on reward-wise gratification. It enables the emergence of music through the interaction of human agents with the system, but not limiting the coherent production to experts in musical theory and performance. Hence, this approach enables novices to compose interesting musical pieces through the use of imagetic elements in a virtual environment, that turns the experience in a kind of ludic musical instrument. We conducted preliminary experiments with a population of musical experts and non-experts, showing that novices were able to learn and create high-quality musical productions through our visual notation system.

4.2 Research Questions

Some specific problems we are aiming to address throughout this research related on how well our approach can tackle specific questions, such as:

- Is meta-interactivity efficient to extend users’ proficiencies?
- Can music generated by novices through this approach match or even surpass music produced by experts?

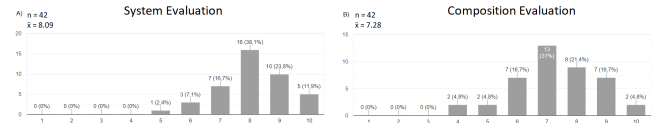


Figure 1: The Bubble Sounds interface evaluation (left) and the evaluation of music produced through it (right).

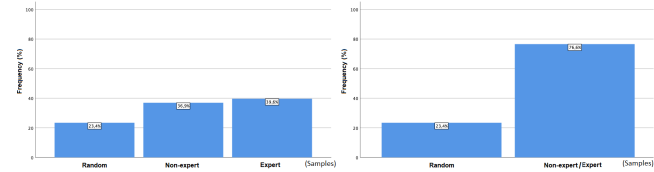


Figure 2: Comparison between the perception of different scenarios of Bubble Sounds experience.

- In meta-interactivity, can visual interfaces combined with sound interfaces create powerful experiences capable to improve the quality of outcomes generated by partially-autonomous systems?
- Does users have control over their own creative process?

4.3 Evaluation

Qualitative and quantitative analysis are being conducted in this research, and statistical tests are being performed (chi-square tests, for our categorical variables, and t-tests, for our numerical variables).

We evaluate our system/approach on its compositional capacity with human subjects, as well as the music produced through it, showing that novice users in musical theory and practice were able to learn and create music as good as experts.

4.4 Current Results

We ran experiments in 2 different stages. In the first stage (n=42) users had an experimentation with Bubble Sounds. In the second stage (n=111), users just listened to the music produced by the system. We also had executions from an autonomous agent performing actions in Bubble Sounds. The experiments we ran with a population of musical experts and non-experts who had an experience with Bubble Sounds showed that non-experts were able to learn and create high-quality musical productions through Bubble Sounds, an example of a meta-interactive approach. As shown in Figure 1, users evaluated both their experience with the system as well as the music they produced through it.

For instance, in our quantitative analysis we performed several analyses over the compositions of users produced during the assessment stage, where we made music produced through our system available for a general public to evaluate the quality of the musical samples. The results, as briefly presented in Figure 2 (left), show an interesting trend of our approach in allowing novices to produce music almost as good as experts. In addition, it also shows that our system is effective on fostering coherent musical corpus even in the chaotic random executions, since a considerable share of the pool acknowledged the random production as good music. It did not, however, dissociate the necessity of the human factor from the relation with the system, as shown in Figure 2 (right), where we present

the result of both novices and experts creation against the random executions. Thus, our algorithm showed to be strong as a device for emergent music, however it is not independent from human feeling/sentiment. We can then assume that the musical feeling associate with sentiment is still important in the creative process of Bubble Sounds, as any form of art, if we analyze it through the perception of a ludic musical instrument.

We run a chi-square test to verify the difference between the perception over the musical production of experts and non-experts, and we found a $p > 0.1$, showing that the difference between them was not statistically significant. This is a very good result, since it reinforces that novices were able to produce music not much different and comparable to the expert's production.

We also verified the perception of music created by human agents (novice and experts) vs. synthetic agents (random samples), as shown in Figure 2 (b). Also according to a chi-square test, we found a $p \leq 0.09$, showing statistical significance, what gives us 91% confidence that human production is likely to be acknowledged as better music than the random generated ones. This result helps us to draw a clear distinction of our partially-autonomous approach if compared to other works, such as Roberts et al. [21] and Hang and Raymond [12], which employ a more autonomous fashion for generating music, not necessarily considering the human expressivity on their productive flow. In our case, our goal is to promote an effective human-machine collaboration, where the first may have creative freedom and the second may provide creative support in a non-limiting way.

5 NEXT STEPS

How generic can our approach be? We have an initial validation of the meta-interactive approach in the context of musical composition in our recent paper, where we allowed individuals to compose musical pieces by playing with color elements in a virtual environment. However, a question that arises concerns on how broad this approach can be. After all, is it possible for this translation system to also achieve satisfactory results in other tasks as well? We believe it is, and outlining how this can be implemented is also in the scope of this research.

New system prototype: For our next work after Bubble Sounds, we intend in retrieving a prior topic that is implicit cooperation [14], that allows a complex musical corpus to emerge from a translation of the user's actions within a virtual environment (so, the player is not aware that he/she is composing a music, since they are actually playing the game and having their actions translated into music).

REFERENCES

- [1] Nipun Agarwala, Yuki Inoue, and Axel Sly. 2017. Music composition using recurrent neural networks. *CS 224n: Natural Language Processing with Deep Learning, Spring 1* (2017), 1–10.
- [2] Florent Berthaut, Victor Zappi, and Dario Mazzanti. 2014. Scenography of immersive virtual musical instruments. In *2014 IEEE VR Workshop: Sonic interaction in virtual environments (SIVE)*. IEEE, Minneapolis, MN, USA, 19–24.
- [3] Marcio Cabral, Andre Montes, Gabriel Roque, Olavo Belloc, Mario Nagamura, Regis RA Faria, Fernando Teubl, Celso Kurashima, Roseli Lopes, and Marcelo Zuffo. 2015. Crossscale: A 3D virtual musical instrument interface. In *2015 IEEE Symposium on 3D User Interfaces (3DUI)*. IEEE, IEEE Xplore, Arles, France, 199–200.
- [4] Pablo Samuel Castro. 2019. Performing Structured Improvisations with pre-trained Deep Learning Models. *arXiv preprint arXiv:1904.13285* (2019), 7.
- [5] Mihaly Csikszentmihalyi. 1990. *Flow: The Psychology of Optimal Experience*. Harper & Row, USA.
- [6] N. Davis, C.-Pi. Hsiao, K. Yashraj Singh, L. Li, and B. Magerko. 2016. Empirically Studying Participatory Sense-Making in Abstract Drawing with a Co-Creative Cognitive Agent. In *Proceedings of the 21st International Conference on Intelligent User Interfaces*. Association for Computing Machinery, Sonoma, California - USA.
- [7] Darina Dicheva, Christo Dichev, Gennady Agre, and Galia Angelova. 2015. Gamification in education: A systematic mapping study. *Journal of Educational Technology & Society* 18, 3 (2015).
- [8] Mark d'Inverno and Jon McCormack. 2015. Heroic versus Collaborative AI for the Arts. In *Proceedings of the 24th International Joint Conference on Artificial Intelligence*. IJCAI 15, Buenos Aires, Argentina.
- [9] Andromeda Entertainment. 2020. SoundSelf: A Technodelic.
- [10] William W Gaver, John Bowers, Andrew Boucher, Hans Gellerson, Sarah Pennington, Albrecht Schmidt, Anthony Steed, Nicholas Villars, and Brendan Walker. 2004. The drift table: designing for ludic engagement. In *CHI 04 - Extended Abstracts on Human Factors in Computing Systems*. ACM, New York, NY, United States, 885–900.
- [11] Simon Holland, Katie Wilkie, Paul Mulholland, and Allan Seago. 2013. Music interaction: understanding music and human-computer interaction. In *Music and human-computer interaction*. Springer, London, UK, 1–28.
- [12] Allen Huang and Raymond Wu. 2016. Deep Learning for Music. *ArXiv abs/1606.04930* (2016), 8.
- [13] Mikhail Jacob and Brian Magerko. 2015. Interaction-based Authoring for Scalable Co-creative Agent. In *Proceedings of the Sixth International Conference on Computational Creativity*. ICCCI, Utah, USA, 236–246.
- [14] Mário Escarce Junior, Georgia Rossmann Martins, Leandro Soriano Marcolino, and Yuri Tavares dos Passos. 2017. Emerging Sounds Through Implicit Cooperation: A Novel Model for Dynamic Music Generation. In *Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*. AAAI, Snowbird, Little Cottonwood Canyon, Utah, USA, 186–192. <https://aaai.org/ocs/index.php/AIIDE/AIIDE17/paper/view/15887>
- [15] Hiroyuki Kotani. 2007. Patapon.
- [16] Teemu Mäki-Patola, Juha Laitinen, Aki Kanerva, and Tapio Takala. 2005. Experiments with virtual reality instruments. In *Proceedings of the 2005 conference on New interfaces for musical expression*. New Interfaces for Musical Expression (NIME05), Vancouver, BC, Canada, 11–16.
- [17] Masaya Matsuura. 1999. Vib Ribbon.
- [18] Chase Mitchusson. 2020. *Indeterminate Sample Sequencing in Virtual Reality*. Ph.D. Dissertation. Louisiana State University – LSU, USA. <https://doi.org/10.5281/zenodo.4813332>
- [19] Julian Moreira, Pierre Roy, and François Pachet. 2013. Virtualband: Interacting with Stylistically Consistent Agents. In *Proceedings of the 14th International Society for Music Information Retrieval Conference*. ISMIR, Curitiba, Brazil.
- [20] François Pachet, Pierre Roy, Julian Moreira, and Mark d'Inverno. 2013. Reflexive loopers for solo musical improvisation. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery – ACM, New York, NY, United States, 4.
- [21] Adam Roberts, Jesse Engel, Curtis Hawthorne, Ian Simon, Elliot Waite, Sageev Oore, Natasha Jaques, Cinjon Resnick, and Douglas Eck. 2017. Interactive musical improvisation with Magenta. In *Proceedings of the Thirtieth Annual Conference on Neural Information Processing Systems*. NIPS, Barcelona, Spain. (Demonstration).
- [22] Katja Rogers, Matthias Jörg, and Michael Weber. 2019. Effects of Background Music on Risk-Taking and General Player Experience. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*. Association for Computing Machinery – ACM, New York, NY, United States, 213–224.
- [23] Katja Rogers, Maximilian Milo, Michael Weber, and Lennart E Nacke. 2020. The Potential Disconnect between Time Perception and Immersion: Effects of Music on VR Player Experience. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*. Association for Computing Machinery – ACM, Virtual Event Canada, 414–426.
- [24] Stefania Serafin, Cumhur Erkut, Juraj Kojs, Niels C. Nilsson, and Rolf Nordahl. 2016. Virtual reality musical instruments: State of the art, design principles, and future directions. *Computer Music Journal* 40, 3 (2016), 22–40.
- [25] Inc. Survios. 2018. Survios.
- [26] Koray Tahiroğlu, Thor Magnusson, Adam Parkinson, Iris Garrelfs, and Atau Tanaka. 2020. Digital Musical Instruments as Probes: How computation changes the mode-of-being of musical instruments. *Organised Sound* 25, 1 (2020), 64–74.
- [27] Ubisoft. 2011. Rocksmith.
- [28] Zhuoming Zhou, Elena Márquez Segura, Jared Duval, Michael John, and Katherine Isbister. 2019. Astaire: A Collaborative Mixed Reality Dance Game for Collocated Players. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*. Association for Computing Machinery – ACM, New York, NY, United States, 5–18.